JOULE · END-PRODUCT SPECIFICATION

Joule Coach

What the finished product looks like — and what stands between the validated engine on GitHub and that product today.

Prepared	10 August 2026
Source repo	github.com/Arslan7111/joule (private) — verified at commit de2a3e7
Companion source	Joule_Coach_Strategy.pdf — product spec + technical roadmap
Method	Every engine claim below was re-checked directly against the cloned repository (src/config.py, src/dashboa
Status	Descriptive target-state document. Sections marked ENGINE TODAY reflect code that exists now; section

1 · Reading this document

How to use what follows

This is a description of **Joule Coach as a finished, shippable product** — the member app, the gym-floor hardware experience, the operator console, and the business wrapped around them — followed by a grounded account of the engineering distance between that end state and the code sitting in the repository today. It merges two sources: the product/technical planning already done in *Joule_Coach_Strategy.pdf*, and a fresh read of the actual **github.com/Arslan7111/joule** codebase (commit **de2a3e7**, ~3,900 lines across **pipeline.py** and 18 modules in **src/**) to confirm nothing below overstates what exists.

> **Nothing described in Part A has been built as a product yet.** The engine that makes it possible — detection, tracking, pose, rep counting, form scoring, 3D fusion — is real, running code, already benchmarked against public datasets (RepCount, CMU Panoptic, Fit3D). The app, backend, persistence, billing, and compliance layer around it are not. Part B states precisely where that line sits, file by file.

Document structure

- **Part A — The end product.** Joule Coach as a member, a gym owner, and floor staff would actually experience it once fully built: the visit end to end, the full feature set, the app screen by screen, the real-time and gamified layer (points, streaks, badges), brand and design language, the business model, and the underlying system architecture.
- **Part B — Distance from today's engine.** A verified, file-by-file account of what exists in the repo right now versus what Part A requires, organized by the same phase structure (F–L) as the technical roadmap, so the gap is auditable rather than asserted.
- **Part C — What could still go wrong.** The honest risk list that determines what an early version of this product is actually allowed to promise a paying member.

PART A

The End Product

Joule Coach as a member, owner, and staff member would experience it once fully built — the product Part B has to earn.

A.1 · The pitch

What Joule Coach is

Joule Coach turns a gym's existing ceiling cameras into a personal trainer that never clocks out. A member checks in with a badge tap, trains as normal, and within seconds of finishing a set sees an honest read of what they did — how many reps, how deep, how controlled, and the one or two things to fix next time — without a phone held up, a wearable, or a human standing over them. Their plan for the week lives in the app; whether they actually did it is measured, not self-reported. For the gym, it is a premium tier members will pay more for, sold and operated without hiring more coaching staff.

What makes this credible rather than another AI-fitness pitch: the computer-vision engine underneath it is not a mockup. It already detects people, tracks them, estimates 3D pose from multiple synchronized cameras, counts reps with a per-person adaptive state machine, and scores form — and every one of those components has already been benchmarked against public research datasets (RepCount, CMU Panoptic, Fit3D), not just demoed on a clean solo video. Part B documents exactly how far that engine already reaches, and where the product wrapped around it still needs to be built.

A.2 · The member experience

A visit, start to finish

01 · Check in

A quick badge tap or QR scan at a wall-mounted kiosk near the entrance. No phone required to start. The kiosk confirms the member's identity is bound to today's visit within about three seconds; if it can't confidently match, a staff member resolves it in two taps rather than the app silently guessing.

02 · Pick the movement

Before a set, the member picks their exercise from a short list on their phone — squat, deadlift, curl, row, overhead press, lateral raise at launch. Anything the system can't yet score (sit-ups, head-on bench press, anything near a mirror) is visibly marked "coming soon," never hidden and never silently mis-scored.

03 · Train, with one glanceable cue

During the set the phone (held or set on a bench) shows just a rep counter and a single light/tone cue if the member drifts out of the camera's frame. No busy live overlay competing for attention mid-lift — that is a deliberate choice, not a missing feature (see A.5).

04 · The report, seconds later

Within a handful of seconds of racking the weight, a report slides up: rep count, per-rep depth validity, a plain-language form read, and the one or two biggest deviations to fix — "knees tracking inward on reps 4–6." Every report states outright whether it came from a 3D fused read (higher confidence) or a single camera (lower), shown as a visual language, not a hedge buried in copy.

05 · The week, tracked honestly

The plan the member built lives in the app; adherence is measured against what the cameras actually saw, not what the member says they did. A "planned vs. done" view distinguishes a genuinely skipped set from a set the cameras simply couldn't see — a camera outage or an uncovered machine is never reported as a missed workout.

06 · Progress that compounds

Trend charts across weeks and months — form-score trajectory, joint-angle history, streaks — framed simply for a first-timer and exposed as raw, exportable data for a member who wants it.

A.3 · The complete feature set

What ships, for whom

The strategy document staged this across MVP → V1 → V2+ sprints; collapsed into the finished product, this is the full surface Joule Coach presents once mature.

For the member

Feature	What it does
Badge/QR check-in	Binds a member's identity to a fresh visit at a fixed kiosk camera; failed matches route to a two-tap staff override, never a silent misattribution.
Guided session start	Closed exercise picker scoped to what the engine can currently score; unsupported movements are labeled, not hidden.
Post-set form report	Rep count, per-rep validity, plain-language form score with 1–2 deviation cues, and an explicit 3D-vs-2D confidence label on every report.
Stay-in-frame cue	A low-latency light/tone signal, independent of and faster than the scoring pipeline, when the member's feet or body leave the calibrated volume.
Plan builder	Weekly plan assembled from a supported-exercise library; blocks adding a movement the member's home gym has no camera coverage for.
Verified adherence	Cross-references the plan against actually-detected sessions — distinguishes "not detected" (camera/coverage issue) from "detected but under-volume," and only the latter may ever read as a missed workout.
Progress dashboard	Multi-session trend charts, persona-adaptive: a simple monthly summary by default, a raw per-joint-angle export mode for advanced members.
Video library	Reference-vs-common-mistake clips mapped 1:1 to exercises the live system actually scores — never a demo for a movement the tracker can't confirm.

Feature	What it does
Injury-risk trend cards	Fatigue, ROM degradation, tempo collapse, and asymmetry surfaced as directional trend cards, not a diagnosis — sustained flags route to a human reviewer.
Reminders	Session and plan nudges that never assert a missed session with more confidence than the underlying detection actually has.

For the gym owner and floor staff

Feature	What it does
Operator console	Camera/station health (last-frame timestamp, calibration-drift warning), member roster, today's session volume; a camera dark for 2+ minutes pages staff.
Usage & business analytics	Station utilization by hour, adherence rates across the membership, churn-risk flags — the number a gym owner actually renews on.
Staff exception panel	Today's failed check-ins, dark cameras, ambiguous identity matches — the human backstop made concrete, not a support-ticket queue.
Install compliance tooling	A guided camera-placement checklist tied to the engine's known blind spots (mirrors, feet-in-frame) with a live warning if a station's frame includes an uncalibrated reflective surface.
Subscription tier gating	Base vs. premium (adds human-reviewer escalation and deeper analytics) — gates feature visibility, never gates how honest a report is.

The human backstop: Remote Coach

The one deliberately-added role that keeps the "no human trainer needed" promise honest rather than reckless: a lightweight remote reviewer queue for the small fraction of cases that genuinely need a person — ambiguous form calls, injury-risk escalations, premium plan customization. It converts the engine's real, named blind spots (bench press needs a side camera, mirrors are unreadable, sit-ups aren't counted yet) into a monetizable premium feature instead of a silent failure the member discovers on their own.

A.4 · The app in your pocket

Screen by screen

The member app is the daily surface for all of the above. Six screens carry almost the entire experience:

Today — The home screen: today's planned session pulled from the member's plan, current check-in state, and a compact snapshot of level, streak, and points — enough to glance at on the way into the gym, nothing that demands attention.

Live Session — Opened once a set starts. Large, glanceable rep counter; the stay-in-frame cue; and — new in this layer — points ticking up in real time as each rep is confirmed (see A.5). No live form-correction overlay here by design; that arrives post-set.

Post-Set Report — Slides up automatically once the set ends: the skeleton overlay for the set's worst rep, plain-language form read, 1–2 corrective cues, and any bonus points/badge progress earned from clean-form reps in that set.

Plan Builder — A calendar-week grid assembled from the supported-exercise library; each card shows its own confidence/coverage badge so a member never plans around a movement the gym can't track.

Progress & Achievements — Trend charts (form-score trajectory, joint-angle history) sit alongside the gamified layer — level, badge case, streak calendar, and an opt-in gym leaderboard (A.5).

Video Library — Reference-vs-common-mistake clips, one per supported exercise, deep-linked straight from a report's corrective cue.

A.5 · Real-time feedback and the gamified layer

Points, streaks, and what "live" honestly means

Two different things get called "real-time" in a product like this, and the end product keeps them clearly separated rather than blurring them into one marketing claim.

Layer	Status	What's actually true
Live rep & points feedback	AVAILABLE NOW	The rep-counting state machine already counts a rep the instant it completes — it's how the engine works today, confirmed in src/state_machine.py and already streamed live over the existing /data endpoint. Gamifying that stream (a live rep counter, a points tick, a streak pulse) is a UI layer on top of something the engine genuinely does in real time — not a promise ahead of the technology.

Layer	Status	What's actually true
Live corrective form feedback	GATED	A mid-rep overlay that tells a member "straighten your back" while they're still in the rep is a different, much harder claim — it needs the full pose→angle→form pipeline running at rep speed with a proven latency budget on real hardware. That validation work is Phase E/H in Part B and has not run yet. Until it has, corrective feedback stays post-set, seconds after the rep — still faster and more specific than a human trainer glancing over occasionally, just not mid-rep.

The points system

Points are earned only from camera-confirmed activity — the same evidentiary standard as adherence tracking (A.3): nothing is awarded from a self-reported set, and nothing is ever deducted for a session the cameras simply didn't see.

Action	Points	Basis
Rep completed	+1	Live, the instant the state machine confirms it — no waiting on form scoring.
Clean-form rep (in the top confidence band)	+3 bonus	Applied retroactively once the post-set form score computes; only on a 3D-fused, high-confidence read.
Full planned set completed	+10	Requires the prescribed reps observed, not just attempted.
Weekly plan fully completed	+50	Same four-state observability model as adherence — only an observed-complete week counts.
Consecutive-day streak	Multiplier, no cap on the downside	Streaks add a multiplier on top of points earned; a missed or unobservable day pauses the multiplier — it is never reset to zero and never triggers a guilt notification.

Levels, badges, and the leaderboard

- **Per-exercise levels** (e.g. Bronze/Silver/Gold Squat) are earned from a rolling trend in consistency and volume, never from a single rep or a single lucky high-confidence read — the same anti-fake-precision principle that governs the form score itself.
- **Badges** are milestone-based and independently verifiable ("100 tracked squats," "4-week streak") and, like every other number in this product, only count what the cameras actually observed.
- **Leaderboard** is opt-in and gym-local by default, ranked on consistency/streak rather than raw weight moved, so it can't quietly push someone toward an unsafe attempt to climb it. A gym can disable it entirely for its floor.

> **The one rule that doesn't bend for gamification:** every anti-pattern already set for the rest of the product applies here too — no streak-shaming, no guilt notifications for a missed or unobservable session, no fake precision in how a point or badge is earned. Points make honest tracking more satisfying to look at; they don't get to lower the bar for what counts as "observed."

A.6 · Look and feel

Design language

"Calibrated confidence, not hype." The single biggest design risk this product carries is overclaiming certainty it doesn't have — a member told "perfect form!" on a rep the system was actually unsure about, who then gets hurt, is the failure mode that kills the business. Every score, count, or alert carries a visible, low-friction confidence signal: solid outline for a high-confidence 3D fused read, dashed for a single-camera 2D estimate, greyed for insufficient data. The voice is a sharp, honest strength coach, not a gamified fitness app — direct and occasionally warm, never cutesy about safety, and never guilt-shaming a missed session.

Charcoal #1c1f22	Dark surface base
Warm white #f4f2ee	Light surface base
Amber #f5a623	Actionable coaching cues & alerts only
Teal #2a9d8f	"Confirmed good" high-confidence states only
Brick red #c0392b	Risk/injury alerts — restrained, never neon

- **Typography:** a workhorse grotesque with tabular figures for every metric and table; a more characterful display face for headlines and coaching-voice moments.
- **Layout:** dense, data-forward for the progress view and operator console; airy, single-focus screens for in-the-moment gym-floor flows (check-in, post-set report).
- **Signature element:** the real skeleton/joint-angle overlay — derived from actual pose data, not a generic icon — used throughout as the app's visual identity. It's the one piece of branding a copycat can't fake, because it's literally the product's own output.
- **Anti-patterns:** no fake precision ("form score: 94.7%" where the error band doesn't support it — round to bands or tiers instead); no stock-photo gym imagery or SaaS gradient-mesh hero sections; no streak-shaming.

A.7 · The business

How Joule Coach makes money

Two-sided subscription: gyms pay per site (camera install + a per-camera or per-member fee) to offer the product as a premium tier; members pay a subscription — either bundled into gym membership or billed directly — for the app, with a Remote Coach add-on as the premium upsell. The real commercial test is not the engine's benchmark accuracy; it's whether a pilot gym's member retention measurably beats its own historical baseline, and whether a second, independently-sourced gym signs on without a founder in the room.

What it tracks	Why it's the real test
60/90-day member retention vs. the gym's own historical baseline	The actual business case — not engagement in isolation, the delta.
Gym owner renewal after the pilot, then a second independently-sourced gym	The second sale is the test of whether this generalizes past one hand-tuned install.
False-"missed"-session rate in adherence tracking	The costlier error direction — must stay near zero or it reads as accusing paying members.
Per-site unit economics (install cost amortized against subscription revenue)	Depends directly on the unmeasured edge-hardware throughput question in Part B.

A.8 · How it's built

System architecture

Three tiers, with a deliberate boundary: raw video never leaves the gym.

EDGE — one device per gym site

Runs inference on-site: camera ingest, detection, pose, per-person analysis (rep counting, form, risk), and the visit-scoped identity gallery. Emits only events and skeleton data outward — never raw frames.

CONTROL PLANE — cloud, multi-tenant

Authenticated ingest and core APIs, tenant-scoped storage (a small, hot relational store for members/visits/reps distinct from a large, cold store for raw angle series), the plan/adherence engine, billing, notifications, and the exercise-video catalog.

APPS — member, operator, content

The member phone app, the gym operator console, and a CDN-delivered video library, all reading from the control plane's APIs.

> **Why edge, not cloud, for inference:** it keeps the compliance surface small (the cloud never holds video of anyone's body), avoids a sustained multi-tens-of-Mbps upload commitment per site running 12+ hours a day, and means a gym with a flaky internet connection still tracks workouts and syncs later instead of going dark. This is treated as close to irreversible once built around.

Identity, done honestly

The end product never persists what a member looks like. A short-lived visit-scoped appearance match (built from clothing colour and generic features, good for maybe 30 minutes) links tracked bodies together during one visit; a durable member_id, established only by an authenticated check-in, is what survives between visits. No appearance template is ever stored — the single most consequential architectural commitment in the whole product, covered in depth in Part B.4.

PART B

Distance From Today's Engine

What is verified, running code in github.com/Arslan7111/joule right now, versus what Part A requires — checked directly against the repository, commit de2a3e7.

B.1 · Verified engine snapshot

What actually runs today

pipeline.py is a command-line tool, not a service: it builds one camera-processing thread per source, runs until the clip or feed ends, and writes a single JSON file on exit. Re-confirmed directly in the repo for this document:

Component	Verified state (commit de2a3e7)
Detector / tracker	YOLOv8n (src/config.py: DETECTOR_MODEL='yolov8n.pt', DETECTOR_CONF=0.5) + ByteTrack.
Pose	MediaPipe, top-down, one call per detected person per frame (cost scales linearly with people in frame).
Rep counting	Exactly two signals: knee average and elbow average. src/exercise_classifier.py's CLASS_FAMILY maps 'situp' and 'cardio' to family 'none' — confirmed still zero reps by design.
Exercise classes	12 labels defined; per README/STATUS the trained model reliably emits a working subset — hinge/lunge/curl/row remain heuristic-only, not trained-model output.
Identity	src/reid.py: MobileNetV3-small ImageNet features + an HSV clothing histogram (_DEEP_WEIGHT=0.6) — confirmed not a Re-ID-trained model. src/config.py: REID_STALE_SECONDS=1800 (30 min).
Persistence	None. SessionGallery is an in-memory dict rebuilt on every run; one JSON write at process exit; a second run into the same output directory overwrites the first.
HTTP surface	src/dashboard.py: unauthenticated http.server on 0.0.0.0, default port 5000 (confirmed — README's documented port 8080 is stale). Routes confirmed live: /, /snapshot/, /data, /people, /checkin (a state-mutating GET, no auth).
Validation record	RepCount Part-A (MAE 0.742, OBO 22.4% aggregate), CMU Panoptic (3.7°–6.1° mean joint-angle error), Fit3D (5–7° for standing exercises, up to 13.3° for squat/pushup) — all confirmed in STATUS.md as already-run, published-dataset benchmarks.
What's absent	Confirmed: no database, no web framework beyond http.server, no tests/ directory, no CI, no Dockerfile, no per-site configuration mechanism.

Read plainly: the perception engine is real and independently validated. Everything Part A needs to turn that into a product — persistence, identity that survives a restart, an authenticated API, an app, a business layer — does not exist in the repository yet.

B.2 · The four gaps that decide whether this is a product

The hard problems, not the busywork

Gap	Size	Why it's the hard one
Real-gym accuracy under crowding, occlusion, mirrors	Unknown	Never tested on real gym footage. The one adverse data point in the repo: the shipped detector caught both people only 30% of the time on heavily-occluded two-person footage (src/config.py benchmark notes).
Throughput on affordable edge hardware	Unknown	Measured today: ~0.22s/frame for 4 fused views with one person, on a laptop. Pose cost is linear in people. Jetson numbers do not exist yet.
Identity that holds for a full visit on a busy floor	Large	Not "persist the gallery" — the embedding is clothing-dominated and won't match the same person next week. The real problem is holding one check-in-established binding through 90 minutes, equipment occlusion, and dozens of similarly-dressed people.
Privacy, consent, biometric-regulation compliance	Large, non-negotiable	The system is legally clean today only because it persists nothing. Every fix above (durable identity, historical storage) actively removes that property — this is the one gap where doing the engineering work creates the legal exposure.

B.3 · Path from here, phase by phase

The build sequence

Condensed from the technical roadmap's Phase F–L structure, sequenced so nothing irreplaceable — a pilot gym's goodwill, its members' trust — is spent on a foreseeable failure.

Phase	Goal	What it delivers
F	Persistence & identity spine	Wall-clock time, durable session/visit/member records, an append-only event stream replacing "one JSON at exit," bounded memory, and the identity-binding model itself — no gym or real footage required, can start today.
G	Privacy & compliance gate	Consent lifecycle, retention/deletion enforced and tested, access audit, and killing the unauthenticated dashboard outright. Hard gate: no real member's body may be processed commercially before this lands.
H	Always-on floor runtime	Turns the CLI script into a supervised service: RTSP timestamp sync, a measured (not estimated) Jetson throughput number, back-pressure policy, per-site configuration.
I	Backend API + operator console + first pilot gym	Authenticated APIs, check-in as the identity anchor, and the single most important deliverable of the whole roadmap — a human-ground-truth accuracy audit that produces the number marketing is actually allowed to use.
J	Member app: plans, adherence, content	The founder's product vision made real, built only once I's accuracy numbers exist — building this earlier risks designing a feature the engine can't honestly support.
K	Multi-tenant scale-out	Gyms two through N: true tenant isolation, billing, fleet operations, a self-serve site-commissioning workflow.

Phase	Goal	What it delivers
L	Model & coverage expansion	Continuous, parallel track: hip-driven rep counting (unlocks sit-ups), wider trained-class coverage, reference-based absolute form scoring, mirror dead-zone masking.

B.4 · The one architectural decision that can't be undone

Identity: bind at check-in, never persist appearance

This is called out on its own because getting it wrong is not fixable after the fact. The temptation, once a product needs to recognize returning members, will be to "just save the gallery." Confirmed directly from src/reid.py and src/session.py: the embedding is generic ImageNet features plus a clothing-colour histogram, explicitly built around clothing staying constant within one session — it will not reliably match the same person wearing different clothes next Tuesday. Persisting it anyway would be both technically weak and, because a stored appearance template derived from someone's body is the textbook definition of a regulated biometric identifier (BIPA, GDPR Art. 9), the most expensive mistake available in this roadmap.

The end product instead keeps three layers strictly separate: a tracklet (seconds, never persisted), a visit-scoped person (one visit, RAM only, never persisted), and a member (forever, but only an ID and a binding record — never body data — created by an authenticated check-in event, not by appearance matching).

PART C

What Could Still Go Wrong

The honest risk list that determines what an early version of this product is actually allowed to promise a paying member.

C.1 · Full risk register

Four categories, plainly stated

Category	What could go wrong
Accuracy	Rep-count accuracy is not yet a product number (only benchmarked on public datasets, not real gym floors); bench press can't be reliably counted head-on; sit-ups count zero by design; mirrors are a total, unclosable blind spot in a typically mirror-walled gym; feet-out-of-frame degrades silently, not with an error.
Identity & trust	Mis-attribution is the system's own documented "worst failure mode" — as a product it becomes a billing dispute, not just a data error; conservative matching produces fragmentation that looks like data loss; bystanders (staff, guests, minors) are currently enrolled with no consent model.
Platform & legal	The current HTTP dashboard is a live, unauthenticated video feed of the gym floor to anyone on the network; BIPA's private right of action is the single highest-cost legal exposure in the US; a bumped camera silently invalidates calibration with no automatic detection today.
Commercial	Per-site unit economics are unknown until real edge-hardware throughput is measured; the shipped detector (Ultralytics YOLOv8) is AGPL-licensed and needs a commercial-license decision before anything ships.

What an early version can honestly promise

- Automatic counting of squats and knee-driven lower-body work, elbow-driven upper-body work, at properly placed, mirror-free, feet-in-frame stations — subject to a real ground-truth accuracy audit at the pilot site.
- Per-set consistency and control feedback with actionable coaching cues, and true 3D joint angles at 3+ camera fused stations.
- A plan builder and adherence tracking for observed activity, with transparent self-logging for everything cameras can't see.

What it must not promise until the engineering work above closes the gap

- Reliable bench-press counting without a true side camera, or any sit-up/core-flexion counting.
- Coverage of mirror-adjacent areas, or an absolute "correctness %" before reference-signature scoring is validated.
- Injury prediction or any medical claim — risk signals are training-load indicators, nothing more.
- Adherence assertions about activity the system genuinely could not observe.

Sources: Joule_Coach_Strategy.pdf (product spec + technical roadmap, prepared 6 Aug 2026) and a direct read of github.com/Arslan7111/joule at commit de2a3e7 — README.md, STATUS.md, ROADMAP.md, DEVELOPMENT.md, CAMERA_PLACEMENT.md, pipeline.py, and src/config.py, src/dashboard.py, src/reid.py, src/session.py, src/exercise_classifier.py, src/output.py. Compiled 10 August 2026.